Report on

Photocatalyst-free Radical Tandem Cyclization/C-C Bond Formation of *N*-allylbromodifluoroacetamides with 2-Methylthiolated Alkynones to Access substituted Thiochromones

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Aim: To synthesize a hybrid molecule bearing thiochromone and 3,3-difluoro-γ-lactam motifs.

Chemicals required:

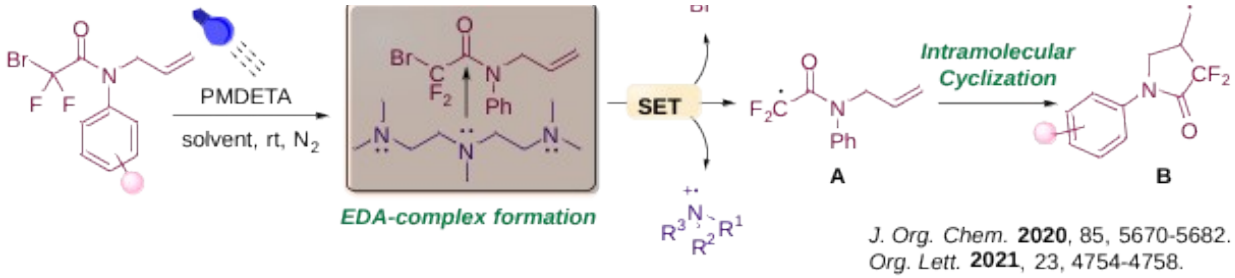
- 2-bromo-2,2-difluoro-*N*-phenylacetamide
- Potassium carbonate
- Allyl bromide
- N,N,N',N",N"-Pentamethyldiethylenetriamine (PMDETA)
- Acetonotrile (CH_3CN)
- Sodium sulfate
- 1-(2-(methylthio)phenyl)-3-phenylprop-2-yn-1-one
- Dimethyl sulfoxide (DMSO)
- Ethyl acetate
- Hexane

Apparatus required:

- Spatula
- 50 mL round-bottomed flask
- Column chromatography setup
- Thin layer chromatography (TLC) plates
- 7 mL screw-cap vial
- Water condenser
- Separating funnel
- Funnel
- Beakers and conical flasks
- Rotatory Evaporator
- Blue LED setup
- Stirring beads

Literature review:

There are a few literature reports on radical tandem cyclization of *N*-allylbromodifluoroacetamides under different photochemical conditions. In these reports, *N*-allylbromodifluoroacetamide forms an electron donor acceptor-complex (EDA-complex) with N,N,N',N",N"-Pentamethyldiethylenetriamine (PMDETA) as a donor. This complex gets excited in visible light and undergoes single electron transfer to form the difluoroalkyl radical **A**. This reactive radical **A** undergoes intramolecular cyclization to form the radical **B**. This radical reacts with substrate molecule and forms substituted 3,3-difluoro-γ-lactam scaffolds.

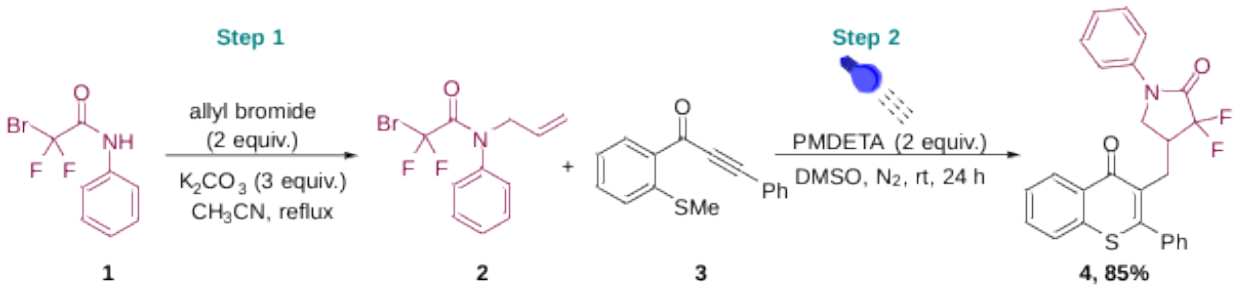


Scheme 1: Formation of difluoroalkyl radical from *N*-allylbromodifluoroacetamide

Procedure:

Step 1: To a solution of 2-bromo-2,2-difluoro-*N*-phenylacetamide **1** (500 mg, 1.0 equiv., 2.0 mmol) in CH_3CN (20 mL), K_2CO_3 (828 mg, 6.0 mmol, 3 equiv.) and allyl bromide (480 mg, 4 mmol, 2.0 equiv.) were added in a 50 mL round-bottomed flask. The mixture was heated to reflux until the reaction was completed as monitored by TLC. The reaction mixture was quenched by adding 30 mL water and was extracted with ethyl acetate (3 X 30 mL). The combined organic layer was dried over Na_2SO_4 and concentrated under reduced pressure. The resulting residue was purified by silica get column chromatography using hexane as an eluent to afford *N*-allyl-2-bromo-2,2-difluoro-*N*-phenylacetamide **2** in 93% yield.

Step 2: To an oven-dried 7 mL screw cap vial equipped with a stirring bead, a mixture of 1-(2-(methylthio)phenyl)-3-phenylprop-2-yn-1-one **3** (40 mg, 0.2 mmol, 1 equiv.), 2-bromo-2,2-difluoro-*N*-phenylacetamide **2** (120 mg, 0.4 mmol, 2 equiv.) and PMDETA (70 mg, 0.4 mmol, 2 equiv.) were added in DMSO (2 mL). The vial was evacuated and filled with nitrogen three times and stirred at room temperature under blue LED for 24 h. After the completion of the reaction (as monitored by TLC), the reaction was quenched with water and extracted with ethyl acetate (3 X 30 mL). The combined organic layers were dried over Na_2SO_4 and concentrated under pressure. The product was purified by silica gel chromatography by using hexane/ethyl acetate (98:2) as eluent to afford the product **4** in 85% yield.



Scheme 2: Synthesis of 3,3-difluoro-4-((4-oxo-2-phenyl-4H-thiochromen-3-yl)methyl)-1-phenylpyrrolidin-2-one

NMR Spectra:

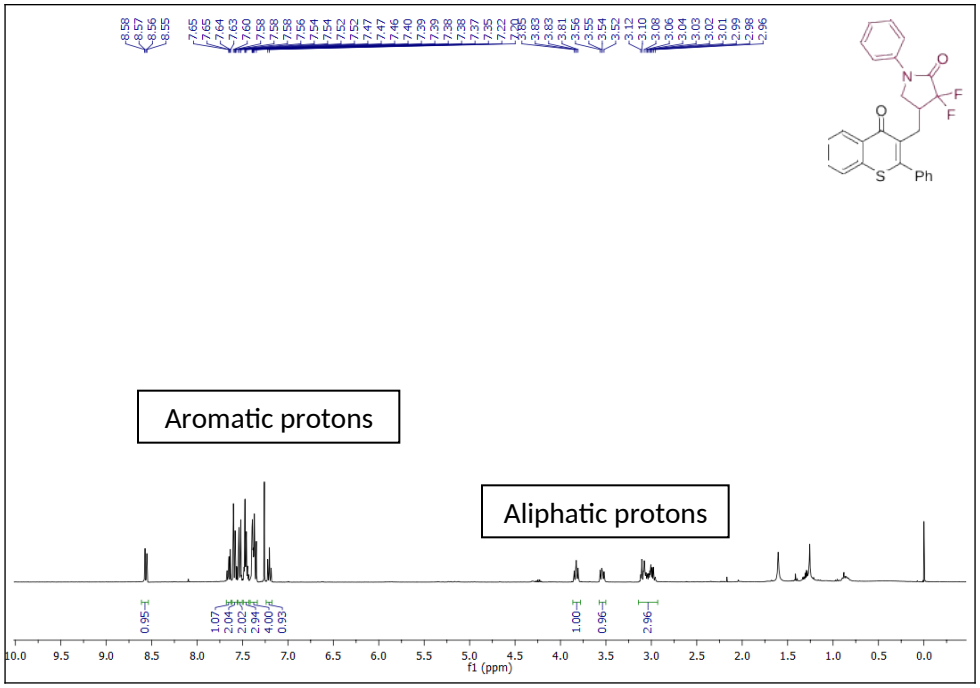


Figure 1: ^{1}H NMR Spectra of **4**

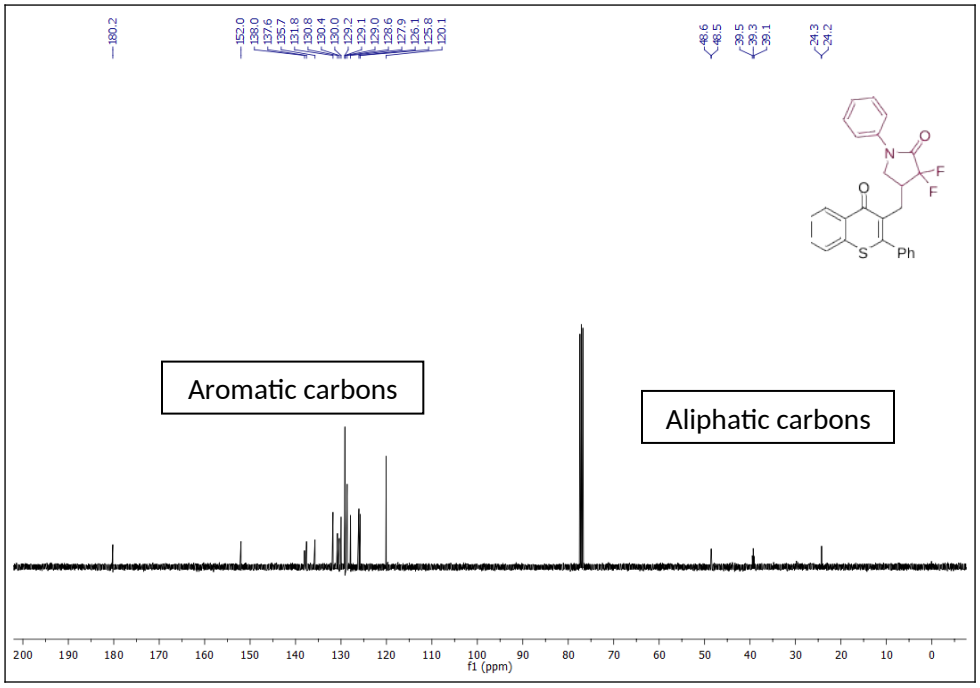


Figure 2: ^{13}C NMR Spectra of **4**

Conclusion:

A colourless solid product **4** was isolated in 85% yield and characterized by NMR spectroscopy.